

ANNEXURE- 'C'

SCHEDULE OF RATES

ANNEXURE C 1.1
SCHEDULE OF RATES FOR SUPPLY, INSTALLATION, TESTING & COMMISSIONING OF MECHANICAL ITEMS /
EQUIPMENTS IN NEW DE-DUSTING UNIT III

Sr. No.	Tag No	Description	Qty.	UOM	Supply (1)		Installation, Testing & Commissioning (2)		Total Value (1) + (2) in Rs.
					Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
1		Compressed Air Tank for Bag Filter (other than Main Receiver by Owner)	1	Nos.					
2		Slide Gate Valves below Hoppers	12	Nos.					
3		Rotary Air Lock Valves	12	Nos.					
4		Screw Conveyor for Bag Filter	7	Nos.					
5	RCF 2000	Recirculation Fan	1	Nos.					
6	FN 2200 & 2300	ID Fan (Free Issue)	2	Nos.	-	-			
7		Motorised Dampers	5	Nos.					
8		Pneumatic Damper	5	Nos.					
9		Manual Damper	14	Nos.					

Sr. No.	Tag No	Description	Qty.	UOM	Supply (1)		Installation, Testing & Commissioning (2)		Total Value (1) + (2) in Rs.
					Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
10		Bag Filter Casing with Inlet & Outlet Plenum, Nozzle House, Pressure Tank, Bag Holding Plate, Bag Cages, Bags, Pneumatic Cylinder for Outlet Damper etc. to complete the System	1	Complete Set					
11		Hot Insulation for Bag Filter & Ducting	1	Lot					
12		Supporting Structure, Operating Platforms, Staircases & Canopy (Pre coated sheets) over the Bag Filter	1	Lot					
13		Misc. Civil works incl. Painting etc in MCC / PLC Room	1	Lot					
TOTAL VALUE									

Note(s):

1. The quantities and Item description is as per the installed System. Bidder shall review & modify as per his System requirements. Any additional Item / Equipment required for completeness of the System shall be brought out by the Bidder and quoted for.
2. After dismantling of the existing System, if any installed installation/Item can be re-used after overhauling/servicing(in Bidder's/Contractor's scope) the same shall be used by the Contractor and the total Contract value shall be reduced by the value of the Item used as per the Schedule of rates given by the Bidder.

ANNEXURE C 1.2
SCHEDULE OF RATES FOR SUPPLY, INSTALLATION, TESTING & COMMISSIONING OF ELECTRICAL ITEMS /
INSTALLATIONS FOR NEW DE-DUSTING UNIT III

Sr. No.	Description	Quantity	UOM	Supply		Install, Test & Commission		Total Value in Rs.
				Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
E1	POWER							
E2	CABLING							
E3	EARTHING							
E4	LIGHTING							
E5	MISCELLANEOUS							
E1	POWER							
E1.1	LT POWER							
E1.1.1	Supply, Installation, Testing & Commissioning of De-dusting MCC panel. The De-dusting MCC panel shall be installed in the De-dusting MCC room.							
	415 V, 300 A, 3 ph. 4-Wire, 50 Hz, 50 KA for 1sec. MCC Panel shall comprise required Feeders: (Existing SLD is attached for reference)	1	Nos.	Free Issue	-			
	(a) Incoming feeder with Loadbreak switch, 315 Amp							
	(b) Outgoing DOL feeders							
	(c) Outgoing SFU feeders							
	(d) Outgoing MCB feeders							
E1.1.2	Supply, Installation, Testing & Commissioning of Electrical Heaters. (The quantity of heaters mentioned is as per existing, however the Bidder's design may require different quantity)	48	Nos.					

Sr. No.	Description	Quantity	UOM	Supply		Install, Test & Commission		Total Value in Rs.
				Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
E1.1.3	Supply, Installation, Testing and Commissioning of Main Heater Junction Box	1	Nos.					
	100A SFU Incomer	1	Nos.					
	32A SFC -Outgoings	13	Nos.					
E1.1.4	Supply, Installation, Testing and Commissioning of Heater JB's (1 no 32A SFU incomer + 5 O/G 16A SFU)	12	Nos.					
	32A SFU Incomer	1	Nos.					
	16A SFU -Outgoings	5	Nos.					
E1.1.5	Supply installation,testing & commissioning of LDB (63A TPN MCB + ELCB, 9 O/Gs of 10A SPN MCBs)	2	Nos.	Free Issue	-			
	63A TPN MCB + ELCB Incomer	1	Nos.					
	10A MCB -Outgoings	5	Nos.					
E1.1.6	Supply installation,testing & commissioning of ELDB having one incomer and two nos. outgoing circuits.	1	Nos.	Free Issue	-			
E1.2	MOTORS							
	Supply Installation, Allignment, Coupling, Teting, Commissioning of weatherproof 3 phase, Induction motor. All erection hardware shall be in Contractor's scope.							
	(a) Horizontal / Vertical mounted induction motor as follows:							
	1) 0.37kW Motor	1	Nos.					
	2) 0.55kW Motor	13	Nos.					
	3) 0.75kW Motor	2	Nos.					
	4) 3.7kW Motor	6	Nos.					
	5) 7.5kW Motor	1	Nos.					
	6) 37kW Motor	1	Nos.					
	7) 500kW Motor (6.6kV)	2	Nos.	-	-			
	The Vendor shall specify the rating as per his System requirements. The above ratings are for existing installation and may be given as Free issue if found in order.							

Sr. No.	Description	Quantity	UOM	Supply		Install, Test & Commission		Total Value in Rs.
				Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
E2.3	CABLE TERMINATIONS							
E2.3.1	Carrying out end termination with weatherproof type cable glands with Nickel plated Brass & blanking plugs, for 1100V grade, Al conductor, PVC insulated, extruded PVC inner sheathed armoured, extruded PVC outer sheathed, Power (Cu / Al) cables of following sizes. These cable glands and tinned Cu. crimping type lugs shall be supplied by Bidder/Contractor. The quantities indicated are to be verified by Bidder.							
	(a) 3C x 4 Sq.mm (Al)	140	Nos.					
	(b) 3C x 6 Sq.mm (Al)	22	Nos.					
	(c) 3C x 10 Sq.mm (Al)	4	Nos.					
	(d) 3C x 70 Sq.mm (Al)	2	Nos.					
	(e) 4C x 16 Sq.mm (Cu)	2	Nos.					
	(f) 4C x 120 Sq.mm (Al)	2	Nos.					
	(g) 3C x 2.5 Sq.mm (Cu)	2	Nos.					
	(h) 3C x 185 Sq.mm	1	Nos.					
	(i) 3.5C x 240 Sq.mm	1	Nos.					
	(j) 3.5C x 300 Sq.mm	1	Nos.					
E2.3.2	Carrying out end termination with weatherproof type cable glands with Nickel plated Brass & blanking plugs, for 6.6kV grade, Al conductor cables. These cable glands and tinned Cu. crimping type lugs shall supplied by Bidder/Contractor. The quantities indicated are tentative and to be verified by Bidder.							
	(k) 3C x 185 Sq.mm (AL) - 6.6kV XLPE/PVC Cable	1	Nos.					
	(l) 3C x 240 Sq.mm (AL) - 6.6kV XLPE/PVC Cable	1	Nos.					
	(l) 3C x 300 Sq.mm (AL) - 6.6kV XLPE/PVC Cable	1	Nos.					

Sr. No.	Description	Quantity	UOM	Supply		Install, Test & Commission		Total Value in Rs.
				Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
E2.3.3	Carrying out end termination with weather proof Nickel plated brass cable glands & Blanking plugs with check nuts, for 1100V grade, Cu conductor, PVC insulated, extruded PVC inner sheathed armoured, extruded PVC outer sheathed, control (Cu) cables of following sizes at LT room end. These cable glands and tinned Cu crimping type lugs shall be supplied by Bidder/Contractor. The quantities indicated are tentative and to be verified by Bidder.							
	(a) 3C x 2.5 Sq.mm (Cu)	24	Nos.					
	(b) 5C x 2.5 Sq.mm (Cu)	48	Nos.					
	(c) 10C x 1.5 Sq.mm (Cu)	8	Nos.					
	(d) 16C x 1.5 Sq.mm (Cu)	10	Nos.					
	(e) 19C x 1.5 Sq.mm (Cu)	12	Nos.					
E2.4	CABLE TRAYS, TRENCH AND ACCESSORIES							
E2.4.1	Supply & Installation of Ladder / Perforated type Cable Tray, pre-fabricated hot dip galvanized Cable trays and other accessories and hardware's shall be in Contractor's scope. All installation hardware's (Nut, Bolt, washers etc.) & required cable tray fixings clamps, etc. shall be in Contractor's scope. Cable trays and accessories required of following sizes (to be verified by Bidder) :							
	a) 600mm Wide Ladder type GI Cable Tray	100	Mtr.					
	b) 300mm Wide Ladder type GI Cable Tray	50	Mtr.					
	c) 300mm Wide Perforated type GI Cable Tray	50	Mtr.					
	d) 150mm Wide Perforated type GI Cable Tray	50	Mtr.					
	e) 600mm Wide Ladder type GI Cable Tray - Horizontal 90 deg Bend	1	Nos.					
	f) 600mm Wide Ladder type GI Cable Tray - Vertical 90 deg Bend	1	Nos.					
	g) 300mm Wide Ladder type GI Cable Tray - Horizontal 90 deg Bend	1	Nos.					
	h) 300mm Wide Ladder type GI Cable Tray - Vertical 90 deg Bend	1	Nos.					
	i) 300mm Wide Perforated type GI Cable Tray - Horizontal 90 deg Bend	1	Nos.					
	j) 300mm Wide Perforated type GI Cable Tray - Vertical 90 deg Bend	1	Nos.					

Sr. No.	Description	Quantity	UOM	Supply		Install, Test & Commission		Total Value in Rs.
				Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
E2.4.2	STEEL MATERIAL							
	Supply, Fabrication, Installation and Painting of structural steel of following sizes for mounting of various equipment in field and for cable tray supporting arrangement. (field equipment like lighting fixture, J.B., receptacles, lighting panels, welding sockets, cable trays etc.) shall be in Contractor's scope. Quantity to be verified by Bidder.							
	a) 50x50x6 mm Angle for cable tray support	100	Kg					
	b) 100x60mm Inset Plate	10	Mtr.					
	c) 6mm Thick Chequered Plate	10	Sq. Mtr.					
	d) ISMC75	200	Kg					
E3	EARTHING & LIGHTNING PROTECTION							
E3.1	Supply & Installation of Earthing conductors shall be as follows. The quantities are tentative and Bidder to verify.							
	a) 10mm dia. GI Wire rope	700	Mtrs.					
	b) 16mm dia. GI Wire rope	30	Mtrs.					
	c) 40 x 5 mm GI Strip	100	Mtrs.					
	d) 25 x 6 mm GI Strip	100	Mtrs.					
E3.2	Testing of Existing Earth pits to achieve proper earthing resistance values. Quantity may vary during execution							
	65mm dia., 3mtr. Long GI pipe as earth electrode.	1	Nos.					

Sr. No.	Description	Quantity	UOM	Supply		Install, Test & Commission		Total Value in Rs.
				Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
E4	LIGHTING							
E4.1	LIGHTING FITTINGS							
E4.1.1	Supply, Installation, Testing and Commissioning of Weather Proof type lighting fittings (including fixing of lamps), pole mounted suitable for outdoor area in Plant. The cable glands shall be part of lighting fixtures. The end termination along with all installation hardware shall be included in Contractor's scope. All lighting fixtures shall be supplied alongwith the lamps. The quantities indicated are tentative and to be verified by Bidder. These fixtures include emergency lighting also.							
	a) Integral wellglass luminaire suitable for 1 x 70 W HPSV lamp.	12	Nos.					
	b) Street light type luminaire suitable for 1 x 150 W HPSV lamp.	12	Nos.					
	c) 36W FTL Lamp replacement in MCC room	10	Nos.					
	d) 18W FTL Lamp replacement in MCC room	5	Nos.					
	e) 18W T5 Lamp replacement in MCC room	5	Nos.					
	f) 70W SV Lamp replacement in existing fixture (this may include repair/replacement of control gear circuit)	12	Nos.					
	g) 150W SV Lamp replacement in existing fixture (this may include repair/replacement of control gear circuit)	12	Nos.					
E4.2	LIGHTING CABLE Supply, Laying & Testing of 1.1KV grade, Cu conductor, PVC insulated, PVC sheathed, GI round wire armoured Lighting cables to be laid in GI perforated type cable trays on pipe sleepers / pipe rack / structure / local support, conduits etc., of following sizes. Installation hardware such as cable nylon ties, clamps, insulation tape, markers, nuts, bolts, washers, screws etc. are in Contractor's scope.							
	(a) 3C x 2.5 Sq.mm (Cu)	500	Mtrs.					
E4.3	JUNCTION BOX Supply, Installation, Testing and Commissioning of weatherproof light junction boxes with canopy. The cable termination with necessary hardware are included in Contractor's scope. The cable glands shall be part of JBs.							
	(a) Junction box - 4 way entry (suitable for 3C-2.5SQMM Cu. Cable)	20	Nos.					

Sr. No.	Description	Quantity	UOM	Supply		Install, Test & Commission		Total Value in Rs.
				Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
E5	MISCELLANEOUS							
	(a) Installation of Shock treatment charts with glass and teak wood frame conforming to IE rules	1	Nos.					
	(b) Installation of first aid wooden box with front door having glass and first aid material / accessories. Contents shall conform to IE rules.	1	Nos.					
	(c) Installation of rubber mats in front of Panels (width 1 meter)							
	i) For service voltage : 415 Volts	10	Mtr.					
	(d) Installation of CO ₂ fire extinguishers (10kg) for Substation Switchgear room / Electrical room.	2	Nos.					
	(e) Refurbishment of Pressurization system for Substation.	1	Lot	Free Issue				
TOTAL OF E1 TO E5								

Note(s):

1. The quantities and Item description is as per the installed System. Bidder shall review & modify as per his System requirements. Any additional Item / Equipment required for completeness of the System shall be brought out by the Bidder and quoted for.
2. After dismantling of the existing System, if any installed installation/Item can be re-used after overhauling/servicing(in Bidder's/Contractor's scope) the same shall be used by the Contractor and the total Contract value shall be reduced by the value of the Item used as per the Schedule of rates given by the Bidder.

ANNEXURE C 1.3
SCHEDULE OF RATES FOR DISMANTLING OF INSTRUMENTATION ITEMS FROM EXISTING INSTALLATION OF
DE-DUSTING UNIT III

Sr. No.	Description	Unit	Qty	Unit Rate in Rs.	Total Value in Rs.
1	Pressure Gauge - Bourdon Type for Bag Filter Unit Connection 1/2" NPT (M), Direct on line mount with Syphon	Nos.	12		
	Temp. Element with Thermowell, Connection 1 1/2" 150# RF				
2	RTD with TW, U length (22 mm- 4 nos. & 1450mm-1 nos)	Nos.	5		
	Thermocouple with TW, U length (975 mm)	Nos.	2		
3	Temperature Transmitters field mounted type	Nos.	3		
4	Pressure Switch, Blind, On Line Connection 1/2" NPT (F), Bellow Type	Nos.	1		
5	Diff. Pressure Switch, Blind, On Line Connection 1/2" NPT (F), Bellow Type	Nos.	1		
6	Level Switch RF type (Capacitance) with Remote electronic switch unit, Process Connection 2" 150# RF, probe length -750 mm	Nos.	12		
7	Pressure Transmitter Pressure Transmitter with 1/2" NPT(F) with 3-way manifold	Nos.	2		
8	Limit Switches for Bag Filter Unit 1" NPT Cable entry, Power supply: 24VDC, 5A, Surface Mount, Contact: 1NO + 1NC (SPDT)	Nos.	24		
	Solenoid Valves				
9	For Damper Valve -2 Port/3-Way	Nos.	2		
	For Power Cylinder -5 Port/4-Way	Nos.	12		
	For Bag Filter Unit -2 Port/3-Way	Nos.	216		

Sr.No.	Description	Unit	Qty	Unit Rate in Rs	Total Value in Rs.
10	Power cylinders actuator for damper with all accessories	Nos.	2		
11	Rotary actuator for dampers with all accessories	Nos.	2		
12	I/P converter with Air Filter Regulator	Nos.	4		
13	TUBE SS / Copper				
	1/4" OD, SS 316 / Copper	Mts	160		
	1/2" OD, SS 316	Mts	30		
14	Cables (armoured / unarmoured)				
	Size : 3C x 1.5mm ²	Mts	3600		
	Size : 5C x 1.5mm ²	Mts	250		
	Size : 7C x 1.5mm ²	Mts	50		
	Size : 10C x 1.5mm ²	Mts	225		
	Size : 19C x 1.5mm ²	Mts	550		
15	Junction Box				
	Pneumatic JB for Piston Cylinder SOVs Size 450x350 mm	Nos	5		
	JB for for Piston Cylinder Limit Switches Size 600x200 mm	Nos	6		
	JB for Bag Filter Unit Size 700x400 mm	Nos	4		

Sr. No.	Description	Unit	Qty	Unit Rate in Rs.	Total Value in Rs.
16	Perforated Cable Trays				
	Size: 50 mm / 100 mm / 150 mm	Mtrs	1		
	Size: 200 mm / 300 mm	Mtrs	1		
17	Local Control Panel for Motorized Inlet Damper Actuator for ID Fans with Temperature Scanner and Panel mounted accessories. Panel of 650 mm x 480 mm and 1150 mm Height.	Nos.	2	Free Issue	-
18	PLC Panel	Nos.	1	Free Issue	-
TOTAL VALUE FOR ITEMS FROM SR. NOS. 1 TO 16					

Note(s):

1. The quantities are indicative and the Bidder shall be paid as per the actual quantity dismantled based on the Unit rates provided.
2. It is envisaged that Items at Serial Nos. 17 & 18 are not required to be dismantled. This needs to be confirmed by the Bidder prior to Offer submission.

ANNEXURE C 1.4
SCHEDULE OF RATES FOR SUPPLY OF INSTRUMENTATION ITEMS FOR
NEW DE-DUSTING UNIT III

Sr. No.	Description	Unit	Qty.	Unit Rate in Rs.	Total Value in Rs.
1	Pressure Gauge - Bourdon Type for Bag Filter Unit Connection 1/2" NPT (M), Direct on line mount with Syphon 4" Dial Size	Nos.	12		
2	Temp. Element with Thermowell, Connection 1 1/2" 150# RF	Nos.	5		
	RTD (Pt-100) with TW, SS316 drills from Barstock				
	U length 22 mm- 4 nos. & fabricated TW SS316 for				
	U length 1450mm -1 nos				
	T/C type K with fabricated TW SS316, U length (975 mm)	Nos.	2		
3	Temperature Transmitters field mounted type for RTD	Nos.	3		
4	Pressure Switch, Blind, On Line (Bellow Type) Connection 1/2" NPT (F), Bottom	Nos.	1		
5	Diff. Pressure Switch, Blind, On Line (Bellow Type) Connection 1/2" NPT (F),	Nos.	1		
6	Level Switch RF type (Capacitance) with Remote electronic switch unit, Process Connection 2" 150# RF,	Nos.	12		
	Probe length 750 mm				
	Pressure Transmitter				
7	Pressure Transmitter with 1/2" NPT(F) 3- way manifold -100 to 0 Pascal - 2 Nos, -5 Kpa to 0 Kpa - 1 Nos				
8	Limit Switches for Bag Filter Unit	Nos.	24		
	Hermetically sealed Microswitch 1 NO, 1 NC contacts				
	Cable entry 1/2" NPT(F)				
9	Solenoid Valves				
a	For Damper Valve -2 Port/3-Way	Nos.	2		
b	For Power Cylinder -5 Port/4-Way	Nos.	12		
c	For Bag Filter Unit -2 Port/3-Way	Nos.	216		
10	Power cylinders actuator for damper with 2 limit Switches.	Nos.	2		
11	Rotary actuator for dampers with all accessories	Nos.	2		
12	I/P converter with Air Filter Regulator	Nos.	4		
13	Instrument Tubing SS / Copper seamless type				
a	Seamless Tube 1/4" OD x 0.036" thk, Copper PVC jacketed	Mtr.	250		
b	Seamless Tube 1/2" OD x 0.049" thk, SS-316	Mts	30		
14	Cables (Armoured type) PVC insulated copper conductor multi stranded FRLS Type				
a	Size : 3C x 1.5mm ² (Screen)	Mts	300		
b	Size : 3C x 1.5mm ²	Mts	3300		
c	Size : 5C x 1.5mm ² (Screen)	Mts	50		
d	Size : 5C x 1.5mm ²	Mts	200		
e	Size : 7C x 1.5mm ²	Mts	50		
f	Size : 10C x 1.5mm ²	Mts	225		
g	Size : 19C x 1.5mm ²	Mts	550		
15	GI Hot dipped Perforated Cable Tray with side coupler plates & SS Nut & Bolts with washer.				
a	Size : 300 mm (W) x 50 mm (H) x 2.5 mm (T)	Mts	40		
b	Size : 200mm (W) x 50 mm (H) x 2.5 mm (T)	Mts	30		
c	Size : 150 mm (W) x 50 mm (H) x 2.5 mm (T)	Mts	50		
d	Size : 100 mm (W) x 30 mm (H) x 2.5 mm (T)	Mts	100		
e	Size : 50 mm (W) x 30 mm (H) x 2.5 mm (T)	Mts	150		
16	Junction Box weatherproof to IP-65 with cable glands, Brass plated double compression type with PVC hood				
a	Pneumatic JB for Piston Cylinder SOVs Size 450 x 350 mm	Nos	5		
b	JB for for Piston Cylinder Limit Switches Size 600 x 200 mm	Nos	6		
c	JB for Bag Filter Unit Size 700 x 400 mm	Nos	4		
17	PLC based monitoring and controlling system which includes Panels, operator station, Printer, all hardware accessories and with software configuration with software licenses as required.	Set	1	Partly Free Issue	
18	Local Control Panel for Motorized Inlet Damper Actuator for ID Fans with Temperature Scanner and Panel mounted accessories. Panel of 650 mm x 480 mm and 1150 mm Height.	Nos.	2	Free Issue	

Note(s):

1. The above Items and quantities are indicative and as per the existing Bag Filter. Bidder to review the Items & quantities as per his scheme for Instrumentation & control in line with the P&ID attached in the Tender Specifications. Required corrections/changes are to be incorporated by the Bidder while submitting the Offer.
2. In case the existing Instruments/Items are used after servicing (in Contractor's scope) , the Contract value will be reduced as per the Unit rate of the Item/Instrument used as mentioned in the Schedule of Rates provided by the Bidder.
3. For Item no. 17, the Owner intends to use the existing PLC Panel & Marshalling Panel. Bidder to confirm prior to Offer submission if he wishes to use /replace the same with justification. However in case the same System is to be used the Bidder shall consider in his scope of supply the Operator Station along with Printer and furniture.
4. For Item no. 18, the Owner intends to use the existing Panels. Bidder to confirm prior to Offer submission if he intends to use or replace the same.

ANNEXURE C 1.5
SCHEDULE OF RATES FOR INSTALLATION, CALIBRATION, TESTING, LOOP
CHECKING & COMMISSIONING OF INSTRUMENTATION FOR NEW DE-
DUSTING UNIT III

A. INSTRUMENTS & HARDWARES

Sr. No.	Description	Unit	Qty	Unit Rate in Rs.	Total Value in Rs.
1	Pressure Gauge - Bourdon Type for Bag Filter Unit Connection 1/2" NPT (M), Direct on line mount with Syphon 4" Dial Size	Nos.	12		
2	Temp. Element with Thermowell, Connection 1 1/2" 150# RF RTD (Pt-100) with TW, SS316 drilled from Barstock				
	U length 22 mm- 4 nos. & fabricated TW SS316 for	Nos.	5		
	U length 1450mm -1 nos				
	T/C type K with fabricated TW SS316, U length (975 mm)	Nos.	2		
3	Temperature Transmitters field mounted type for RTD	Nos.	3		
4	Pressure Switch, Blind, On Line (Bellow Type) Connection 1/2"	Nos.	1		
5	Diff. Pressure Switch, Blind, On Line (Bellow Type) Connection	Nos.	1		
6	Level Switch RF type (Capacitance) with Remote electronic switch	Nos.	12		
7	Pressure Transmitter				
	Pressure Transmitter with 1/2" NPT(F) 3- way manifold -100 to 0	Nos.	3		
8	Limit Switches for Bag Filter Unit Hermetically sealed Microswitch	Nos.	24		
9	Solenoid Valves				
a	For Damper Valve -2 Port/3-Way	Nos.	2		
b	For Power Cylinder -5 Port/4-Way	Nos.	12		
c	For Bag Filter Unit -2 Port/3-Way	Nos.	216		
10	Power cylinders actuator for damper with 2 limit Switches.	Nos.	2		
11	Rotary actuator for dampers with all accessories	Nos.	2		
12	I/P converter with Air Filter Regulator	Nos.	4		
13	Instrument Tubing SS / Copper				
a	Seamless Tube 1/4" OD x 0.036" thk, Copper PVC jacketed	Mtr.	250		
b	Seamless Tube 1/2" OD x 0.049" thk, SS-316	Mts	30		
14	Laying of armoured Cables including Dressing, Clamping, including supply of Al clamps, Roofing nut & bolts for clamping, SS cable tags etc..				
a	Size : 3C x 1.5mm ² (Screen)	Mts	300		
b	Size : 3C x 1.5mm ²	Mts	3300		
c	Size : 5C x 1.5mm ² (Screen)	Mts	50		
d	Size : 5C x 1.5mm ²	Mts	200		
e	Size : 7C x 1.5mm ²	Mts	50		
f	Size : 10C x 1.5mm ²	Mts	225		
g	Size : 19C x 1.5mm ²	Mts	550		
15	GI Hot dipped Perforated Cable Tray with side coupler plates & SS Nut Bolt with washer.				
a	Size : 300 mm (W) x 50 mm (H) x 2.5 mm (T)	Mts	40		
b	Size : 200mm (W) x 50 mm (H) x 2.5 mm (T)	Mts	30		
c	Size : 150 mm (W) x 50 mm (H) x 2.5 mm (T)	Mts	50		
d	Size : 100 mm (W) x 30 mm (H) x 2.5 mm (T)	Mts	100		
e	Size : 50 mm (W) x 30 mm (H) x 2.5 mm (T)	Mts	150		
16	Junction Box weatherproof to IP-65 with cable glands, Brass plated double compression type with PVC hood				
a	Pneumatic JB for Piston Cylinder SOVs Size 450 x 350 mm	Nos	5		
b	JB for for Piston Cylinder Limit Switches Size 600 x 200 mm	Nos	6		
c	JB for Bag Filter Unit Size 700 x 400 mm	Nos	4		
17	PLC based monitoring and controlling system which includes Panels, operator station, Printer, all hardware accessories and with software configuration with software licenses as required.	Set	1		
18	Local Control Panel for Motorized Inlet Damper Actuator for ID Fans with Temperature Scanner and Panel mounted accessories. Panel of 650 mm x 480 mm and 1150 mm Height.	Nos.	2		

B. LAYING OF CABLES & TRAYS, FIXING SUPPORTS, CABLE DRESSING AND CLAMPING, GLANDING & TERMINATION, INSTALLATION OF JUNCTION BOXES, TUBING WORKS ETC.

Sr. No.	Description	Unit	Qty.	Unit Rate in Rs.	Total Value in Rs.
1	Armoured Cables including Dressing, Clamping, Glanding & Termination along with supply of Al clamps, Roofing nut & bolts for clamping , SS cable tags etc..				
a	Size : 3C x 1.5mm ² (Screen)	Mts	300		
b	Size : 3C x 1.5mm ²	Mts	3300		
c	Size : 5C x 1.5mm ² (Screen)	Mts	50		
d	Size : 5C x 1.5mm ²	Mts	200		
e	Size : 7C x 1.5mm ²	Mts	50		
f	Size : 10C x 1.5mm ²	Mts	225		
g	Size : 19C x 1.5mm ² (Screen)	Mts	550		
2	Earthing cables for grounding of Insts, JB & Panels, Unarmoured type PVC insulated multi-stranded copper conductor wire in green colour				
a	1 x 2.5 MM ² multistrand copper cable, PVC insulated unarmoured type in green colour.for JB & Field insts	Mtr.	30		
b	1 x 6 MM ² multistrand copper cable, PVC insulated unarmoured type green colour for Analyzer Transmitter	Mtr.	30		
c	1 x 25 MM ² multistrand copper cable, PVC insulated unarmoured type in green colour for DCS Control Panel	Mtr.	30		
3	Tubing work with including fittings, Valves with Proper supports & clamping & including supply of GI 'U' clamps, nut & bolts as reqd.				
a	Seamless Tube 1/4" OD x 0.036" thk, Copper	Mtr.	250		
b	Seamless Tube 1/2" OD x 0.049" thk, SS-316	Mts	30		
4	Servicing of Junction Box including wiring termination checking, cleaning terminals of the JB. Replacement of defective terminals.				
a	Pneumatic JB for Piston Cylinder SOVs	Size 450 x 350 mm	Nos	5	
b	JB for for Piston Cylinder Limit Switches	Size 600 x 200 mm	Nos	6	
c	JB for Bag Filter Unit	Size 700 x 400 mm	Nos	4	

Sr. No.	Description	Unit	Qty.	Unit Rate in Rs.	Total Value in Rs.
5	Fabrication & Erection of GI Hot dipped Perforated Cable Tray including fabrication of bends, reducers, Tees and supply of mounting GI nut & bolts.				
a	Size : 300 mm (W) x 50 mm (H) x 2.5 mm (T)	Mtr.	40		
b	Size : 200mm (W) x 50 mm (H) x 2.5 mm (T)	Mtr.	30		
c	Size : 150 mm (W) x 50 mm (H) x 2.5 mm (T)	Mtr.	50		
d	Size : 100 mm (W) x 30 mm (H) x 2.5 mm (T)	Mtr.	100		
e	Size : 50 mm (W) x 30 mm (H) x 2.5 mm (T)	Mtr.	150		
6	Fabrication, Erection & epoxy Painting of structural Steel like Angle, Channels Flats, Plate. For support of Cable Trays, Panels, JB, Air Manifold & Field Inst.	kg	1000		
7	Fabrication, Installation & epoxy Painting of 2" Instrument Stand Pipe (Stanchion support) including supply of 4 nos Anchor Fasteners, 300x300 mm, 6 mm thick base plate. Pipe shall be 2" size IS :1239 hvy duty.	Mtr.	30		
8	Laying of 1/2" pipe with Argon Welding work including fittings, Valves with Proper supports & clamping & with supply of 'U' clamps, nut & bolts, welding Electrodes as reqd.				
a	Pipe 1/2" NB, SCH. 80 , ASTM A 106	Mtr.	80		
b	Pipe 1/2" NB, SCH. 40S , SS 316	Mtr.	20		

Note(s):

1. The above quantities are based on the existing installed System. Bidder to review the Items & quantities based on his System requirement.
2. For Item no. 17 & 18 Bidder to note the comments given in Annexure C 1.4.

ANNEXURE C 1.6
SCHEDULE OF RATES FOR TESTING, SERVICING & CALIBRATION OF DISMANTLED INSTRUMENTATION ITEMS OF
EXISTING DE-DUSTING UNIT III

Sr.No.	Description	Unit	Qty	Unit Rate in Rs.	Total Value in Rs.
1	Pressure Gauge - Bourdon Type for Bag Filter Unit Connection 1/2" NPT (M), Direct on line mount with Syphon	Nos.	12		
2	Temp. Element with Thermowell, Connection 1 1/2" 150# RF				
	RTD with TW, U length (22 mm- 4 nos. & 1450mm-1 nos)	Nos.	5		
	Thermocouple with TW, U length (975 mm)	Nos.	2		
3	Temperature Transmitters field mounted type	Nos.	3		
4	Pressure Switch, Blind, On Line Connection 1/2" NPT (F), Bellow Type	Nos.	1		
5	Diff Pressure Switch, Blind, On Line Connection 1/2" NPT (F), Bellow Type	Nos.	1		
6	Level Switch RF type (Capacitance) with Remote electronic switch unit, Process Connection 2" 150# RF, probe length -750 mm	Nos.	12		
7	Pressure Transmitter Pressure Transmitter with 1/2" NPT(F) with 3- way manifold	Nos.	2		
8	Limit Switches for Bag Filter Unit 1" NPT Cable entry, Power supply: 24VDC, 5A, Surface Mount, Contact: INO + INC (SPDT)	Nos.	24		
9	Solenoid Valves				
	For Damper Valve -2 Port/3-Way	Nos.	2		
	For Power Cylinder -5 Port/4-Way	Nos.	12		
	For Bag Filter Unit -2 Port/3-Way	Nos.	216		

Sr. No.	Description	Unit	Qty	Unit Rate in Rs.	Total Value in Rs.
10	Power cylinders actuator for damper with all accssories	Nos.	2		
11	Rotary actuator for dampers with all accssories	Nos.	2		
12	I/P converter with Air Filter Regulator	Nos.	4		
13	TUBE SS / Copper				
	1/4" OD, SS 316 / Copper	Mts	160		
	1/2" OD, SS 316	Mts	30		
14	Cables (armoured / unarmoured)				
	Size : 3C x 1.5mm ²	Mts	3600		
	Size : 5C x 1.5mm ²	Mts	250		
	Size : 7C x 1.5mm ²	Mts	50		
	Size : 10C x 1.5mm ²	Mts	225		
	Size : 19C x 1.5mm ²	Mts	550		
15	Junction Box				
	Pnuematic JB for Piston Cylinder SOVs Size 450 x 350 mm	Nos	5		
	JB for for Piston Cylinder Limit Switches Size 600 x 200 mm	Nos	6		
	JB for Bag Filter Unit Size 700 x 400 mm	Nos	4		
16	Local Control Panel for Motorized Inlet Damper Actuator & ID Fan with Temperature Scanner and Panel mounted accessories. Panel of 650 mm x 480 mm and 1150 mm Height.	Nos.	2		
TOTAL VALUE					

Note(s):

1. The Unit rates will be applicable in case the existing installed Items are found serviceable on joint inspection by the Owner & Contractor after dismantling. In case a particular Item or Instrument is used (after servicing by Contractor), the Contract value for supply of Instrumentation Item will decrease by the Unit rate provided in Annexure C 1.4 and the Contractor shall be paid as per the Unit rates given in this SOR (Annexure C 1.6)
2. For Testing, Servicing / Repairs & Calibration the amount payable shall be for the actual numbers / quantities tested, serviced and calibrated based on the Unit rates quoted.
3. All parts/components/hardware etc. required for the servicing /overhauling shall be in the scope of supply of the Contractor.

ANNEXURE C 1.7

SCHEDULE OF RATES FOR ITEMS NOT CONSIDERED IN SCOPE /REQUIRED DURING PROJECT EXECUTION

Sr. No.	Tag No	Description	Qty.	UOM	Supply (1)		Fabrication, Installation, Testing & Commissioning (2)		Total Value (1) + (2) in Rs.
					Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
1		Inlet Duct (CS) of Size 1200 mm Dia.	1	Meter					
2		Inlet Duct (CS) of Size 3000 mm Dia.	1	Meter					
3		Inlet Duct (CS) of Size 3200 mm Dia.	1	Meter					
4		Castable Lining for 1200 mm Dia. Duct	1	Meter					
5		Miter Bend for 1200 mm Dia. Duct	1	Nos.					
6		Miter Bend for 3000 mm Dia. Duct	1	Nos.					
7		415V, 300 Amp, 3 Ph, 4-Wire, 50 Hz, 50 KA for 1 Sec. MCC Panel	1	Nos.					
8		Light Distribution Board (63 A TPN MCB + ELCB, 9 O/Gs of 10A SPN MCBs)	1	Nos.					
9		Emergency Light Distribution Board consisting of One Incomer and Two Outgoing Circuits	1	Nos.					
10	Power Cables								
		3Cx4 Sq.mm Al conductor, PVC insulated, PVC sheathed, GI round wire armoured cable (69 Nos.)	1060	Meter					
		3Cx6 Sq.mm Al conductor, PVC insulated, PVC sheathed, GI round wire armoured cable (11 Nos.)	200	Meter					

Sr. No.	Tag No	Description	Qty.	UOM	Supply (1)		Fabrication, Installation, Testing & Commissioning (2)		Total Value (1) + (2) in Rs.
					Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
		3Cx10 Sq.mm Al conductor, PVC insulated, PVC sheathed, GI round wire armoured cable (2 Nos.)	55	Meter					
		3Cx70 Sq.mm Al conductor, PVC insulated, PVC sheathed, GI round wire armoured cable (1 No.)	60	Meter					
		4Cx120 Sq.mm Al conductor, PVC insulated, PVC sheathed, GI round wire armoured cable (1 No.)	25	Meter					
		4Cx16 Sq.mm Cu conductor, PVC insulated, PVC sheathed, GI round wire armoured cable (1 No.)	30	Meter					
		3Cx2.5 Sq.mm Cu conductor, PVC insulated, PVC sheathed, GI round wire armoured cable (1 No.)	35	Meter					
11	Control Cables								
		3Cx2.5 Sq.mm Cu conductor, PVC insulated, PVC sheathed, GI round wire armour cable (12 Nos.)	120	Meter					
		5Cx2.5 Sq.mm Cu conductor, PVC insulated, PVC sheathed, GI round wire armour cable (24 Nos.)	24	Meter					

Sr. No.	Tag No	Description	Qty.	UOM	Supply (1)		Fabrication, Installation, Testing & Commissioning (2)		Total Value (1) + (2) in Rs.
					Unit Rate in Rs.	Total Value in Rs.	Unit Rate in Rs.	Total Value in Rs.	
12	Cables between MCC and PLC								
		10Cx1.5 Sq.mm Cu conductor, PVC insulated, PVC sheathed, GI round wire armour cable (4 Nos.)	70	Meter					
		16Cx1.5 Sq.mm Cu conductor, PVC insulated, PVC sheathed, GI round wire armour cable (6 Nos.)	90	Meter					
		19Cx1.5 Sq.mm Cu conductor, PVC insulated, PVC sheathed, GI round wire armour cable (6 Nos.)	120	Meter					
13		Local Control Panel for Motorized Inlet Damper Actuator for ID Fans with Temperature Scanner and Panel mounted accessories. Panel is of 650 mm x 480 mm and 1150 mm Height.	2	Nos.					
14		New PLC based Control System as per Article No. 5.4.3.2	1	No.					
TOTAL VALUE									

Note(s):

1. The quantities and Item description is as per the installed System.
2. The Unit rates provided by the Bidder shall be applicable when the Free Issue Item considered by the Owner is found damaged /non-functional and needs to be replaced by the Contractor additionally.

ANNEXURE C 2.0

SCHEDULE OF PRICE (BREAK-UP OF TOTAL PRICE) FOR DISMANTLING, DESIGN & ENGINEERING, SUPPLY, INSTALLATION, TESTING & COMMISSIONING OF NEW DE-DUSTING UNIT-III

Serial No.	Description	Total Value in Rs.
1	Lump-sum Value for Dismantling of existing De-dusting Unit III & Refurbishment / overhauling as defined under Article 4.0 (Except for Instrumentation Items as mentioned in Annexure C 1.3)	
2	Design & Engineering of New De-dusting Unit III	
3	Total Value of Schedule of Rates for Supply, Installation, Testing & Commissioning of Mechanical Items/Equipments for New De-dusting Unit III as per Annexure C 1.1	
4	Total Value of Schedule of Rates for Supply, Installation, Testing & Commissioning of Electrical Items/Installations for New De-dusting Unit III as per Annexure C 1.2	
5	Total Value of Schedule of Rates for Dismantling of Instrumentation Items from existing Installation of De-dusting Unit III as per Annexure C 1.3	
6	Total Value of Schedule of Rates for Supply of Instrumentation Items for New De-dusting Unit III as per Annexure C 1.4	
7	Total Value of Schedule of Rates for Installation, Calibration, Testing, Loop Checking & Commissioning of Instrumentation for New De-dusting Unit III as per Annexure C 1.5	
8	Minimum guaranteed CENVAT credit on account of Service Tax/ Excise	
	Total Contract Price (net of CENVAT)	

Note(s):

1. Bidder is responsible to verify the Items & quantities as mentioned in various Schedule of Rates.
2. Values to be mentioned above shall include all Taxes and Duties as applicable.
3. In case, any job from Annexure- C 1.6 and C- 1.7 is required, the qualifying L1 party as per Annexure- C 2.0 may be asked to supply that item at the lowest of the rates pertaining to that item as quoted by all the bidders.